

FULL DETAILS (Read-only) -> [Click Here to Create PDF for Current Dataset of Trial](#)

CTRI Number	CTRI/2020/01/022821 [Registered on: 16/01/2020] Trial Registered Prospectively															
Last Modified On:	16/11/2020															
Post Graduate Thesis	No															
Type of Trial	Observational															
Type of Study	Case Control Study															
Study Design	Other															
Public Title of Study	Whole exome sequencing of patients having comorbidities associated with type 2 diabetes.															
Scientific Title of Study	Identification of risk alleles associated with Diabetes Kidney Disease (DKD) using the whole exome sequencing approach															
Trial Acronym	ZiDA															
Secondary IDs if Any	<table border="1"> <thead> <tr> <th>Secondary ID</th><th>Identifier</th></tr> </thead> <tbody> <tr> <td>NIL</td><td>NIL</td></tr> </tbody> </table>	Secondary ID	Identifier	NIL	NIL											
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Details of Principal Investigator or overall Trial Coordinator (multi-center study)	<table border="1"> <tr> <td>Name</td><td>Dr Ratnesh Bhai Mehta</td></tr> <tr> <td>Designation</td><td>Head Genome Informatics</td></tr> <tr> <td>Affiliation</td><td>Zifo RnD Solutions</td></tr> <tr> <td>Address</td><td>Temple Steps 2nd Floor, Block 2, Near Little Mount Metro Station, Saidapet Chennai TAMIL NADU 600015 India</td></tr> <tr> <td>Phone</td><td></td></tr> <tr> <td>Fax</td><td></td></tr> <tr> <td>Email</td><td>ratnesh.m@zifornd.com</td></tr> </table>		Name	Dr Ratnesh Bhai Mehta	Designation	Head Genome Informatics	Affiliation	Zifo RnD Solutions	Address	Temple Steps 2nd Floor, Block 2, Near Little Mount Metro Station, Saidapet Chennai TAMIL NADU 600015 India	Phone		Fax		Email	ratnesh.m@zifornd.com
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Source of Monetary or Material Support	<table border="1"> <tr> <td>Zifo RnD Solutions (Self Sponsored)</td></tr> </table>	Zifo RnD Solutions (Self Sponsored)																							
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Sites of Study Modification(s)	<table border="1"> <tr> <th colspan="4">No of Sites = 10</th></tr> <tr> <th>Name of Principal Investigator</th><th>Name of Site</th><th>Site Address</th><th>Phone/Fax/Email</th></tr> <tr> <td>Dr A J Asirvatham</td><td>Arthur Asirvatham Hospital</td><td>42-a, Kuruvikaran Salai, Anna bus stand, Madurai, Tamil Nadu 625020 Madurai TAMIL NADU</td><td>04522535266 ajasirvathamresearch@yahoo.in</td></tr> <tr> <td>Dr R Gobinath</td><td>Dr. Gobinath Diabetes Specialty Centre</td><td>38133, Anna Nagar East, 2nd Main Road, Anna Nagar, Madurai -625020 Madurai TAMIL NADU</td><td>4524514135 rrg1954@gmail.com</td></tr> <tr> <td>Dr Melvin George</td><td>Hindu Mission Hospital</td><td>103, GST Road, Tambaram, Chennai, Tamil Nadu- 600045 Chennai TAMIL NADU</td><td>9894133697 melvingeorge2003@gmail.com</td></tr> <tr> <td>Dr K Narayanan</td><td>Kidney Center Hospital</td><td>#206, Anna Salai, Near old bus stand.</td><td>8754038971 drnana@gmail.com</td></tr> </table>	No of Sites = 10				Name of Principal Investigator	Name of Site	Site Address	Phone/Fax/Email	Dr A J Asirvatham	Arthur Asirvatham Hospital	42-a, Kuruvikaran Salai, Anna bus stand, Madurai, Tamil Nadu 625020 Madurai TAMIL NADU	04522535266 ajasirvathamresearch@yahoo.in	Dr R Gobinath	Dr. Gobinath Diabetes Specialty Centre	38133, Anna Nagar East, 2nd Main Road, Anna Nagar, Madurai -625020 Madurai TAMIL NADU	4524514135 rrg1954@gmail.com	Dr Melvin George	Hindu Mission Hospital	103, GST Road, Tambaram, Chennai, Tamil Nadu- 600045 Chennai TAMIL NADU	9894133697 melvingeorge2003@gmail.com	Dr K Narayanan	Kidney Center Hospital	#206, Anna Salai, Near old bus stand.	8754038971 drnana@gmail.com
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			Pondicherry â€ 605001 Pondicherry PONDICHERRY	
	Dr S Ravikumar	Levin Diabetes Speciality Hospital	313, Karpaga Nagar 11th Street, Karpaga Nagar, K.Pudur, Madurai â€ 625007,Tamil Nadu Madurai TAMIL NADU	8870004595 ravekumar24@gmail.com
	Dr Krishnaswamy Sampathkumar	Meenakshi Mission Hospital & Research Centre (MMHRC)	Department of Nephrology Meenakshi Mission Hospital and Research Centre (MMHRC) Lake Area, Melur Road 36, Radhakrishnan Street, Bibkulam Madurai TAMIL NADU	9994872250 drsampathkumar.k@mmhrc.in
	Dr MC Deepak Lal	R.M DIABETES EDUCATION & RESEARCH FOUNDATION	AA2, 2nd Avenue, Anna Nagar Round Tana, Chennai- 600040, Tamil Nadu Chennai TAMIL NADU	04426211919 drdiabdeepak@gmail.com
	Dr K Subramaniyan	SRM Medical College Hospital & Research Centre	SRM Nagar, Potheri, Kattankulathur -643243 Tamil Nadu, india Chennai TAMIL NADU	9443307525 subramaniyan.35@gmail.com
	Dr Georgi Abraham	The Madras Medical Mission	4-A, Dr. J. Jayalalitha Nagar, Mogappair Chennai TAMIL NADU	9841710992 abraham_georgi@yahoo.com
	Dr Arun Kannan	The Madurai Institute Of Diabetes And Endocrine Practice & Research	East Veli Street, Madurai City, Madurai â€ 625001, Tamil Nadu Madurai TAMIL NADU	94877333563 drarunkannan@gmail.com
Details of Ethics Committee Modification(s)	No of Ethics Committees= 11			

	Name of Committee		Approval Status
	Institutional Ethics Committee Arthur Asirvatham Hospital		Approved
	Institutional Ethics Committee Hindu Mission Hospital		Approved
	Institutional Ethics Committee SRM Medical College Hospital And Research Center		Approved
	Institutional Ethics Committee The Madras Medical Mission		Approved
	JCDC_Dr Arun Kannan		Approved
	JCDC_Dr K Narayanan		Approved
	JCDC_Dr M C Deepak Lal		Approved
	JCDC_Dr R Gobinath		Approved
	JCDC_Dr S Ravikumar		Approved
	Meenakshi Mission Hospital & Research Centre		Approved
	Ripon Independent Ethics Committee		Approved
Regulatory Clearance Status from DCGI	Status		
	Not Applicable		
Health Condition / Problems Studied	Health Type	Condition	
	Healthy Human Volunteers	Participant with past history of Type 2 diabetes	
	Patients	(1) ICD-10 Condition: E112 Type 2 diabetes mellitus with kidney complications,	
Intervention / Comparator Agent	Type	Name	Details
	Comparator Agent	Nil	NIL, This is an observational study
Inclusion Criteria	Age From	18.00 Year(s)	
	Age To	85.00 Year(s)	
	Gender	Both	
	Details	Case and control group will be confirmed based on Blood Fasting Glucose, Hb1AC, UACR, Serum Ceratinine and eGFR	
ExclusionCriteria	Details	No history of Type 1 diabetes No nephropathy unrelated to diabetic nephropathy	
Method of Generating Random Sequence	Not Applicable		
Method of Concealment	Not Applicable		
Blinding/Masking	Not Applicable		
Primary Outcome	Outcome		TimePoints
	1) Identification of biomarkers to study disease progression / pathophysiology		3 to 4 years down the line.

Secondary Outcome	<table border="1"> <thead> <tr> <th>Outcome</th> <th>TimePoints</th> </tr> </thead> <tbody> <tr> <td>1) Person specific tailored effective, safe medicine and dose (pharmacogenomics) 2) Biomarker based early diagnosis (preventive medicine)</td> <td>3 to 4 years down the line</td> </tr> </tbody> </table>	Outcome	TimePoints	1) Person specific tailored effective, safe medicine and dose (pharmacogenomics) 2) Biomarker based early diagnosis (preventive medicine)	3 to 4 years down the line	
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1) Person specific tailored effective, safe medicine and dose (pharmacogenomics) 2) Biomarker based early diagnosis (preventive medicine)	3 to 4 years down the line					
Target Sample Size	Total Sample Size="20000" Sample Size from India="20000" Final Enrollment numbers achieved (Total)= "Applicable only for Completed/Terminated trials" Final Enrollment numbers achieved (India)= "Applicable only for Completed/Terminated trials"					
Phase of Trial	N/A					
Date of First Enrollment (India)	20/01/2020					
Date of Study Completion (India)	Applicable only for Completed/Terminated trials					
Date of First Enrollment (Global)	Date Missing					
Date of Study Completion (Global)	Applicable only for Completed/Terminated trials					
Estimated Duration of Trial	Years="3" Months="0" Days="0"					
Recruitment Status of Trial (Global) Modification(s)	Not Applicable					
Recruitment Status of Trial (India)	Open to Recruitment					
Publication Details	No publications yet					
Individual Participant Data (IPD) Sharing Statement	Will individual participant data (IPD) be shared publicly (including data dictionaries)? Response - NO					
Brief Summary	SUMMARY In India, as per the Global Burden of Disease (GBD) in 2017, chronic kidney disease is the sixth leading cause of morbidity and mortality (2.39% of total deaths reported). Diabetes kidney disease (DKD) is a type of chronic kidney disease caused by diabetes. It is a slow progressive disease that damages the kidneys and incapacitates them from performing their function of filtering the blood like they should, which can cause wastes to build up in the body. The disease can eventually lead to kidney failure, also called as End Stage Renal Disease (ESRD). Diabetes kidney disease is known to be influenced by genetic factors (hereditary information in our DNA present in our cells). Due to the increasing prevalence of DKD and it being a burden on health-care, there has been an increase in the interest for the search of genetic					

factors that are associated with the disease, since it will provide an understanding of the mechanism of disease development and progression, biomarkers and inputs for further research to develop novel drug targets. Identification of these genetic factors in this study will be done using a methodology called Whole Exome Sequencing (WES), which has been performed for various ethnic groups globally (Eur J Hum Genet. 2016 Aug;24(8):1175-80, Front Genet. 2019; 10: 494, Front Genet. 2019 Feb 26;10:118) but there is limited information for the Indian population. The current study is not a regular clinical trial study but more a genotype-phenotype observational study in the line of Exome Wide Association Study (EWAS). The overall objective of this study is to identify novel genes and functional variants associated with Diabetic Kidney Disease in Indian population primarily focusing upon South Indian population. Identification of such novel genes / variants can be used on for the purpose of disease diagnosis and prognosis. The study will involve whole exome sequencing / SNP genotyping among patients with DKD and matched controls (diabetics who don't have DKD).

MATERIALS AND METHODS

Study Participants: Study participant will be recruited from local hospitals /clinics. Approval from institutional ethics committee will be taken for current study. CTRI registration is required for this purpose. All participants will be recruited to the study after written informed consent from them. Based on the eligibility criteria in consultation with clinicians participants will either be placed in the case arm, or the control arm of the study group. Approximately, 20000 ethnic matched (10000 case arm; 10000 control arm) participants will be recruited. The age of the participants both for case and control arm should be between 18 -85 years with South Indian ancestry.

Sequencing and Statistical Analysis: Whole blood (5-8 mL) along with all available clinical (Annexure 1) and medication history (Past & Present) will be collected from the participant by trained technician / phlebotomist at hospital / clinic under medical supervision as a part of whole exome sequencing project. No tissue biopsy or biopsy sample will be collected in the current study. The participants will be double barcoded (pseudonymized) and key to identifiers will be stored in limited accessed password protected computer / server. Current study involves very limited need to know personal identifier.

Briefly, DNA will be isolated either from blood. Exome sequencing will be performed using capture array platform with average coverage of 100x. Multiple level of quality control (QC) will be carried out in respect to samples and sequence data. Reference aligned sequence reads will be filtered based on multiple QC criteria, including number of mapped reads, fraction of properly paired reads, distribution of insertion sizes, distribution of average base quality, as well as guanine and cytosine bias. Variants will be called and annotated as per GATK better practices guideline and ACMG germline variant calling guideline. Association analysis in DKD and non DKD samples for all markers in the dataset in respect to clinical phenotype will be carried out. In addition, multiple association tests will be performed (Allelic association test, Cochran-Armitage trend test, dominant gene action test, recessive gene action test, genotypic test).

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